

Effect of protective grazing shelters on relative humidity and temperature for Ponderosa pine seedlings

Chandler Craft and James Levie

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Abstract

- Summarize the data, findings, and conclusions of the project in 250 words or less.

Introduction

Summary

The increasing variability of precipitation and rising temperature in the Southwest are contributing to increasingly frequent and larger wildfires. This leads to large expanses of severely burned areas, which carry negative complications for monsoon and flood conditions in summer storms, potentially increasing the risk for loss of life or property. Beyond this, these severely burned areas lack the seed sources required for natural regeneration, preventing recovery. Ponderosa pines in particular have a limited seed dispersal distance of 150 feet, further limiting post-fire regrowth. These seedlings have consistently poor survival rates in the Southwest, leading to the use of protective planting shelters to improve survival odds. These assist the seedling both by prevent herbivory from ungulates, rodents, and rabbits after the seedling is planted, as well as by moderating the climate within the planting structure.

These survival impacts have been minimally studied, particularly the microclimate impacts to Ponderosa pine seedlings within the protective shelters. The impact of these shelters on the microclimate surrounding Ponderosa pine seedlings is the primary area of interest for this study. Thus, microclimate data was collected for a variety of sites for both treated and untreated seedlings, with above-ground temperature, above-ground relative humidity, below-ground temperature, and below-ground water content being the primary microclimate variables of interest.

Data Sources and Processing

Using remote sensing techniques, microclimate data was collected every hour from August of 2024 until August of 2025 on both treatments, one with and one without the shelter. Hourly readings were necessary since preliminary data showed little difference in both temperature and humidity averages throughout the whole day. Hourly readings allowed us on the research team to determine metric fluxes at specific times of the day.

According to our previous analyses and preliminary data, the research team needed averages at the hottest and coldest parts of the day across all sensors instead of comparing entire daily averages by treatment.

To clean our data, the “Lubridate” package was primarily used for multiple functions and lent the most help to us on the cleaning side.

First, it parsed a combined time and date code and separated them into separate columns. This would have been relatively simple otherwise, but this dataset consists of hourly readings for almost a year of remote sensor work. There are many unique observations.

Secondly, the parsed codes were then kept in 24hr time code and separated from the date code.

Third, the 24hr time codes were translated into 12 hour codes, and then immediately into monikers (AM or PM) and assigned to the dataset so that analyses could be performed on different windows within the same day that corresponded to the lowest and highest adiabatic (moisture) rates.

A summary of the variables used in analysis is shown below in Table 1.

Table 1: Summary of variables of modeling interest

Variables	Descriptions
Date	Date of observation
Site	Site terrain (steep or flat)
Treatment	Site treatment (sheltered or unsheltered)
Minimum aboveground temperature	Daily minimum aboveground temperature of site, Celsius
Maximum aboveground temperature	Daily maximum aboveground temperature of site, Celsius
Minimum relative humidity	Daily minimum relative humidity of site
Maximum relative humidity	Daily maximum relative humidity of site
Minimum belowground temperature	Daily minimum belowground temperature of site, Celsius
Maximum belowground temperature	Daily maximum belowground temperature of site, Celsius
Minimum water content	Daily minimum water content of site
m=Maximum water content	Daily maximum water content of site

Analysis

- First create plots of the data to show what motivated your hypotheses.
- Descriptions of the methods used to analyze the data.
- Descriptions should include a small discussion of the statistical methods and why they are applicable.
- Should contain how model building was performed and if multiple models were built.
- How did you choose which model or models to use over others?
- This is your chance to show me your understanding of the statistical methodology.
- You are not limited to one type of analysis. Multiple results using well prepared models is encouraged.
- The goal should be to provide a detailed exploration of your data and the choices you made to address your questions and project aims.

Results

- Display important output tables, graphics, and diagnostics,
- Final models and interpretation.
- The interpretation of meaningful variables can be an important result!
- Evaluation of the model and how it relates to your expert opinion.
- After all of the work in the Analysis section was done, what were the major outcomes?

Conclusion

- Discuss the success of the project.
- Did you learn something new that you find interesting?
- What goals and aims have been addressed?
- Is the resulting analysis useful and how so?
- Is something else needed?
- Were any goals left unmet?
- Any new ideas now that you have done the project?

References

- References to any external resources should be included.
- Does not count maximum page count.